- **Duality**
- **Advanced Linear programming**

Find the dual of problems:

Problem 1.
$\min_x 3x_1 - 9x_2$
such that

$$2x_1 - x_2 + x_3 \geq 3,$$
$$x_1 - x_2 - 3x_3 \geq 2,$$
$$x_1, x_2, x_3 \geq 0.$$

Solution

The dual
$\max_x 3y_1 + 2y_2$
such that

$$2y_1 + y_2 \leq 3,$$
$$-y_1 - y_2 \leq -9,$$
$$y_1 - 3y_2 \leq 0,$$

$$y_1, y_2 \geq 0.$$

Problem 2. $\min_x 3x_1 - 9x_2$
such that

$$4x_1 - x_2 + x_3 = 3,$$
$$x_1 - x_2 - 6x_3 \geq 2,$$
$$x_1 - 4x_2 - 6x_3 \leq 8,$$
$$x_1, x_2 \geq 0.$$

Solution

The dual is
$\max_y 3y_1 + 2y_2 - 8y_3$
such that

$$4y_1 + y_2 - y_3 \leq 3,$$
$$-y_1 - y_2 + 4y_3 \leq -9,$$
$$y_1 - 6y_2 + 6y_3 = 0,$$

$$y_2 \geq 0, \quad y_3 \geq 0.$$

Problem 3. Consider the following problem
$\min_x 2x_1 + 9x_2 + 3x_3$
such that

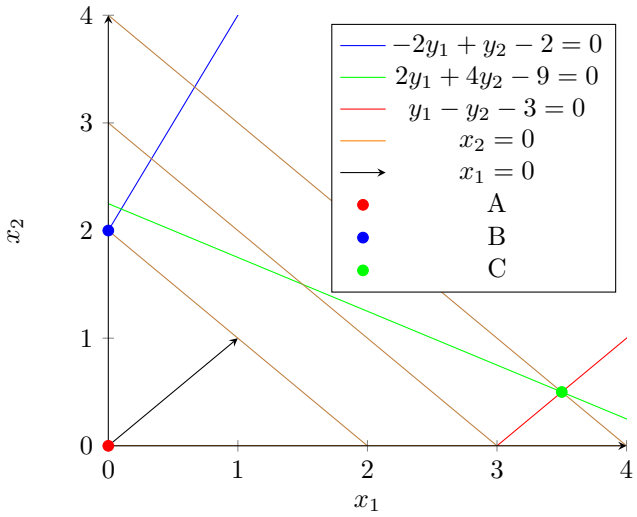
$$-2x_1 + 2x_2 + x_3 \geq 1,$$
$$x_1 + 4x_2 - x_3 \geq 1,$$
$$x_1, x_2, x_3 \geq 0.$$

Find the dual of this problem and solve it graphically. Use complementary slackness conditions to obtain a solution of the primal.

Solution

The dual is
$\max_y y_1 + y_2$
such that

$$-2y_1 + y_2 \leq 2,$$
$$2y_1 + 4y_2 \leq 9,$$
$$y_1 - y_2 \leq 3,$$
$$y_1, y_2 \geq 0.$$



It is clear that the maximum is attained at the green point $(7/2, 1/2)$ and is equal to 4.

Proposition (Complementary Slackness)

If x and y are optimal, then

$$x^T(c - A^T y) = 0 \quad \text{and} \quad y^T(Ax - b) = 0.$$

Equivalently, componentwise:

1. For all i, either $x_i = 0$, or $(A^T y)_i = c_i$ (i.e. $c_i - A^T_{:i} y = 0$).
2. For all j, either $y_j = 0$, or $(Ax)_j = b_j$ (i.e. $A_{j:} x - b_j = 0$).

By the proposition (see Lecture notes) above we have

$$y^T (Ax - b) = 0$$

$$(x_1, x_2, x_3)(c - A^T y)$$

where

$$A = \begin{pmatrix} -2 & 2 & 1 \\ 1 & 4 & -1 \end{pmatrix}$$

$$b^T = (1,1), \quad c^T = (2,9,3), \quad y^T = (7/2, 1/2).$$

Consequently,

$$(7/2, 1/2) \begin{pmatrix} -2x_1 + 2x_2 + x_3 - 1 \\ x_1 + 4x_2 - x_3 - 1 \end{pmatrix} = 0$$

$$(x_1, x_2, x_3) \begin{pmatrix} -2y_1 + y_2 - 2 \\ 2y_1 + 4y_2 - 9 \\ y_1 - y_2 - 3 \end{pmatrix} = 0$$

Since $y^T = (7/2, 1/2)$ we get

$$\begin{gathered} \tfrac{7}{2}(-2x_1 + 2x_2 + x_3 - 1) = 0 \\ \tfrac{1}{2}(x_1 + 4x_2 - x_3 - 1) = 0 \\ x_1(-2 \cdot \tfrac{7}{2} + \tfrac{1}{2} - 2) = 0 \\ x_2(2 \cdot \tfrac{7}{2} + 4 \cdot \tfrac{1}{2} - 9) = 0 \\ x_3(\tfrac{7}{2} - \tfrac{1}{2} - 3) = 0 \end{gathered}$$

It possible when $x^T = (0, 1/3, 1/3)$. Therefore, the optimal solution of the initial problem is $(0, 1/3, 1/3)$ and is equal to 4.

Problem 4. How to solve the problem $\min_x 50x_1 + 25x_2$
such that

$$x_1 + 3x_2 \geq 8,$$
$$3x_1 + 4x_2 \geq 19,$$
$$3x_1 + x_2 \geq 7,$$
$$x_1, x_2 \geq 0,$$

without an initialization phase?

Solution

To begin with we need to write the dual. We have
$\max_y 8y_1 + 19y_2 + 7y_3$
such that

$$y_1 + 3y_2 + 3y_3 \leq 50,$$
$$3y_1 + 4y_2 + y_3 \leq 25,$$
$$y_1, y_2, y_3 \geq 0.$$

Let's transform into a min problem in standard form. We obtain
$\min_y -8y_1 - 19y_2 - 7y_3$
such that

$$y_1 + 3y_2 + 3y_3 + y_4 = 50,$$
$$3y_1 + 4y_2 + y_3 + y_5 = 25,$$
$$y_1, y_2, y_3, y_4, y_5 \geq 0.$$

Use simplex method, we get
Based on the optimal dual solution, derive the optimal primal solution with strong duality and complementary slackness
If $y_1 = y_2 = y_3 0$ then a vertex is $(0, 0, 0, 50, 25)$.
Using the previous equalities, we get

Basic	y_1	y_2	y_3	y_4	y_5	Solution
z	-8	-19	-7	0	0	0
y_4	1	3	3	1	0	50
y_5	3	4	1	0	1	25

The solution (0, 0, 0, 50, 25) is the BFS associated with the basic variables x_4 and x_5.
The reduced costs associated with the variables x_1, x_2 and x_3 are negative.
Since the most negative number in z-line is -19, we choose to enter x_2 in the base.
We can't increase x_2 without limit, since we have to satisfy the constraints

$$\begin{cases} 3x_2 + x_4 = 50, \\ 4x_2 + x_5 = 25. \end{cases}$$

The second constraint is the most restrictive since $\frac{50}{3} \geq \frac{25}{4}$. It's x_5 that leaves the base.
Iteration 1.

Basic	y_1	y_2	y_3	y_4	y_5	Solution
z	25/4	0	-9/4	0	19/4	475/4
y_4	-5/4	0	9/4	1	-3/4	125/4
y_2	3/4	1	1/4	0	1/4	25/4

Iteration 2.
The solution (0, 25/4, 125/4, 0, 0) is the new BFS associated with the basic variables y_2 and y_4.
The non-basic variables are y_1, y_3 and y_5.

The reduced costs associated with the variables y_3 is negative. The cost decreases if y_3 increases.
We choose to enter y_3 in the base. We can't increase x_3 without limit, since we have to satisfy the constraints:
1. The first imposes $9/4y_1 + y_4 = 125/4$.
2. and the second $x_2 + 1/4x_3 = 25/4$.
The first constraint is the most restrictive since $125/9 \geq 25$. It's y_3 that leaves the base.
After elementary transformations on the rows, we obtain the canonical table

Basic	y_1	y_2	y_3	y_4	y_5	Solution
z	5	0	0	1	4	150
y_3	-5/9	0	1	4/9	-1/3	125/9
y_2	8/9	1	0	-1/9	1/3	25/9

The associated reduced costs are positive. Therefore, the solution is optimal. The minimum is attained at $(0, 25/9, 125/9)$ and is equal to -150.

Proposition (Complementary Slackness)

If x and y are optimal, then

$$x^T(c - A^T y) = 0 \quad \text{and} \quad y^T(Ax - b) = 0.$$

Equivalently, componentwise:

1. For all i, either $x_i = 0$, or $(A^T y)_i = c_i$ (i.e. $c_i - A^T_{:i} y = 0$).
2. For all j, either $y_j = 0$, or $(Ax)_j = b_j$ (i.e. $A_{j:}x - b_j = 0$).

Using the Proposition above (see Lecture Notes for more details), then

$$(0, 25/9, 125/9)\,(Ax - b) = 0$$

where

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 4 \\ 3 & 1 \end{pmatrix}$$

$$b^T = (8, 19, 7)$$

$$y^T = (0, 25/9, 125/9).$$

Consequently,

$$(0, 25/9, 125/9) \begin{pmatrix} x_1 + 3x_2 - 8 \\ 3x_1 + 4x_2 - 19 \\ 3x_1 + x_2 - 7 \end{pmatrix} = 0$$

and

$$0 \cdot (x_1 + 3x_2 - 8) = 0,$$

$$25/9(3x_1 + 4x_2 - 19) = 0,$$

$$125/9(3x_1 + x_2 - 7) = 0.$$

Hence,

$$3x_1 + 4x_2 = 19$$
$$3x_1 + x_2 = 7$$

We need to solve this system linear equations. Therefore, the optimal solution of the initial problem is $(1, 4)$ and by strong duality the maximum is equal to 150.